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Final Year Project Report

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Made within ATOMIC IT

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General Introduction

Mobility is one of the most basic conditions of modern economic and social life. Access to reliable transportation determines whether students reach their universities on time, whether workers can take a job in another city, and whether families can maintain regular contact across long distances. In Tunisia, however, this everyday need collides with a transportation system that has not kept pace with the growth of the population or with the geographic distribution of universities, industrial zones, and tourist regions. Public transport is often overcrowded and irregular, fuel prices have risen steadily, and inter-city travel remains expensive and time-consuming for those who do not own a vehicle.

Carpooling has emerged worldwide as a natural answer to this kind of imbalance. By matching drivers who already plan a trip with passengers who are heading in the same direction, it transforms unused seats into a shared resource, reduces the cost of mobility for both sides, and lowers the environmental impact of individual driving. In Tunisia, the demand for such a service is clearly present : thousands of trips between cities such as Tunis, Sfax, Sousse, and the smaller towns of the coast and the interior are organized every week. The supply, however, remains almost entirely informal, scattered across Facebook groups, private messages, and word-of-mouth contacts.

This informal organization carries real consequences. There is no verification of identity, no record of payment, no structured search, and no mechanism to evaluate a driver or a passenger after a trip. International platforms such as BlaBlaCar offer a more structured experience, but their adoption in Tunisia is held back by payment models that do not match local habits, by interfaces that are not localized, and by a user base that is too small to ensure frequent matches on the most requested national routes. The result is a paradoxical situation : a strong demand for shared mobility coexists with the absence of a digital tool truly designed for the country.

It is precisely this gap that the present project addresses. **ForsaDrive** is a carpooling platform conceived from the start for the Tunisian context. It is built as a unified ecosystem composed of a web application, a mobile application, and a shared backend, so that the same business rules apply regardless of the device used. The platform supports the complete lifecycle of a shared trip : account creation, identity and student verification, ride publication, search and booking, in-app communication, payment with a fifty-percent prepayment that confirms the booking, ratings, and complaint handling. It also introduces features that respond to specific local realities, including a verified student status that grants an automatic fifty-percent discount, support for cash payment of the remaining balance, and an interface designed for English, French, and Arabic users. The project was carried out within the company **ATOMIC IT**, which provided the professional environment, the technical supervision, and the methodological framework needed to take the idea from a sketch to a structured deliverable. The remainder of this report is organized into five chapters. The first chapter presents the host organization and the general framework of the project, including a critical analysis of the current transportation situation in Tunisia and the motivations that led to ForsaDrive. The second chapter conducts a detailed analysis of existing solutions, identifies the actors of the system, formalizes the functional and non-functional requirements, and explains the agile methodology adopted during the internship. The third chapter is devoted to the conception of the platform : it describes the general architecture, refines the most representative use cases, presents the dynamic behavior of the system through sequence and activity diagrams, and lays down the structural backbone of the application through a class diagram and a relational schema. The fourth chapter describes the implementation, the technological choices, and the most representative screens of the resulting application. The fifth chapter discusses validation, testing, and deployment. The report closes with a general conclusion that summarizes the contributions of the project and outlines the perspectives for future work.

Chapitre 1

Project Framework

Plan

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Introduction

Before describing the technical work carried out on ForsaDrive, it is necessary to situate the project in its concrete context. This chapter therefore presents the host organization within which the internship was conducted, examines the current state of mobility in Tunisia and the limits of the solutions that users currently rely on, and finally states the motivations and objectives that led to the development of the platform. The chapter closes with an outline of the proposed solution, which serves as the bridge towards the analysis of requirements presented in the next chapter.

1.1 Study of the Host Organization

1.1.1 Presentation of ATOMIC IT

ATOMIC IT is an information technology engineering services company specialized in the design and development of digital solutions. Its team brings together engineers, technicians, and IT specialists who work primarily on web and mobile development. The company assists its clients in the full lifecycle of their software projects, from the initial expression of needs to the deployment and maintenance of the delivered system.

Beyond pure development, ATOMIC IT positions itself as a partner that helps its clients translate business requirements into reliable digital tools. It relies on modern technologies and on programming environments such as Windows and Linux, the .NET and Java ecosystems, and the major mobile platforms (Android and iOS). This breadth of expertise allows the company to work on a wide range of projects, from institutional websites to platforms with significant business logic such as the one developed during this internship.

1.1.2 General Information about ATOMIC IT

Table 1.1 summarizes the main administrative information about the host organization.

Tableau 1.1 : General information about ATOMIC IT

Address	Arab League Street, Kélibia, Nabeul Governorate
Phone Number	+216 55 343 224
Activity	Web and mobile development, software engineering, and digital consulting
Director	Mr. Khalil Selmi
Core focus	Design and delivery of responsive web platforms and mobile applications adapted to client needs

1.1.3 Services Offered by ATOMIC IT

The company structures its activity around four main services that complement each other.

Website Creation. Design of responsive and customized websites, with attention to performance, accessibility, and scalability. The company adapts the technical stack to the client's profile, whether the goal is a content-driven site or a more dynamic interface.

Web Engineering. Development of more complex platforms such as e-commerce systems, community platforms, and data-driven applications. This service covers backend architecture, database modeling, and the integration of third-party components such as payment gateways or notification services.

Web Marketing. Support in the definition of digital marketing strategies. The company assists its clients in the choice of channels, the structuring of campaigns, and the alignment of online communication with measurable business objectives.

Search Engine Optimization. Improvement of the visibility of websites in search engines through keyword strategies, indexing, and the application of best practices in technical and content SEO.

The internship took place primarily within the engineering activity of the company, since ForsaDrive belongs to the family of data-driven platforms with significant business logic.

1.1.4 Organizational Chart of ATOMIC IT

The internal organization of ATOMIC IT is deliberately flat and oriented towards short communication paths between decision making and execution. Figure 1.1 presents the simplified organizational chart of the company.

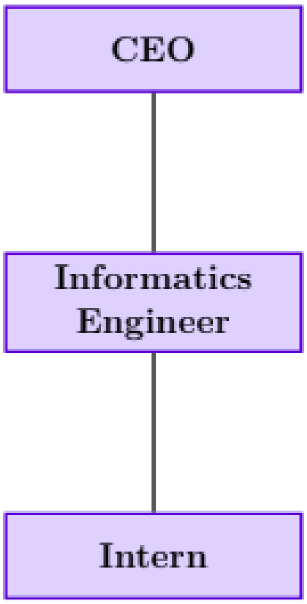

Figure 1.1 : Organizational chart of ATOMIC IT

At the top level, the chief executive officer supervises strategic decisions, validates project directions, and represents the company with its clients. The development teams handle the technical implementation, while interns contribute to ongoing projects under the supervision of senior developers. This structure favors quick feedback loops, which proved beneficial during the internship since the validation of design choices on ForsaDrive could take place without intermediate delays.

1.1.5 Internship Framework

The internship was carried out within the development team of the company, in coordination with both frontend and backend developers. The work followed an agile Scrum approach, with daily stand-up meetings inside the team and weekly reviews with the supervising engineer. This rhythm allowed a continuous validation of progress and gave the opportunity to adjust priorities as the project advanced. The two interns who collaborated on ForsaDrive worked on the same shared application rather than on two independent products. The work was therefore distributed by functional modules and by sprints, not by a strict separation between web and mobile responsibilities. This organization made it possible to keep the business logic consistent across the two interfaces and to share the design effort on the most sensitive parts of the system, such as authentication, the booking flow, and the payment logic.

1.2 Critical Study of the Existing Situation

1.2.1 General Context

Mobility plays a central role in economic activity, education, and social interaction. In Tunisia, transportation conditions every aspect of daily life, from the choice of a university to the feasibility of a job offer in another governorate. Yet, the available means of transportation suffer from structural limitations that make travel both expensive and unpredictable. Public transport between cities is often overcrowded and irregular, the road network does not always serve secondary towns with the same frequency as the main corridors, and private cars travel every day with empty seats that no system currently helps to fill.

This situation places a particular burden on students and young professionals, who travel frequently but have limited budgets. It is in this segment that the demand for a reliable and affordable shared mobility service is the most visible.

1.2.2 Analysis of Current Limitations

The mobility solutions currently available in Tunisia each address part of the problem but leave significant gaps. Public transport remains the cheapest option, but its capacity is limited and its schedule does not match the flexibility that students or workers often need. Informal carpooling, which is mostly organized through Facebook groups and private messaging, fills part of the demand but does so without any structure : there is no verification of the identity of drivers or vehicles, no traceability of payments, no centralized search, and no mechanism to rate or report a user after a trip. International applications such as Bolt or inDrive operate on a ride-hailing model designed for short urban trips, and their pricing logic is not adapted to long-distance carpooling. International carpooling platforms such as BlaBlaCar are technically suitable but suffer from a low local adoption, partly because their payment methods, customer support, and user base are not aligned with the realities of the Tunisian market.

Several recurring weaknesses can be summarized as follows :

- Public transport suffers from delays, overcrowding, and insufficient geographic coverage.
- Existing international platforms are not fully adapted to local realities such as payment methods, identity verification, or regional mobility patterns.
- Informal ride-sharing lacks trust mechanisms such as identity checks, ratings, or secure communication channels.
- Private vehicles are underutilized, with many empty seats during daily and weekly trips.

— No centralized solution offers a structured advantage to students, who form a large part of the inter-city travel demand.

These limitations highlight the absence of a structured, secure, and locally adapted system for shared mobility.

1.2.3 Problem Statement

The analysis of the existing situation can be condensed into a single observation :

> *Tunisia lacks a secure, affordable, and well-structured digital ride-sharing solution that is adapted to local payment habits, available on both web and mobile, and capable of building genuine trust between drivers and passengers.*

The challenge is therefore not to invent a new concept, but to adapt the proven model of carpooling to the specific constraints of the Tunisian context.

1.2.4 Project Motivation

The motivation behind ForsaDrive is at the same time social, economic, and technical. On the social side, the project aims to make mobility more accessible to populations whose budget excludes private taxis and whose schedule is not always compatible with public transport, particularly students. On the economic side, the platform contributes to a better use of an existing resource, namely the empty seats of vehicles already on the road, which translates into shared savings for drivers and passengers. On the technical side, the project offers the opportunity to design a complete software ecosystem, articulating a web application, a mobile application, and a shared backend around a coherent business logic.

1.2.5 Project Objectives

The general objective of the project is the design and the implementation of a complete carpooling platform adapted to the Tunisian context. This general objective is broken down into the following specific objectives :

0. Develop a web platform that supports the management of rides and the administration of the system.
0. Build a mobile application that covers the daily interactions of drivers and passengers.
0. Provide a unified backend that exposes a single set of business rules to both clients.
0. Integrate trust mechanisms based on identity verification, ratings, and complaint handling.
0. Implement a wallet-based payment flow with a fifty-percent prepayment that confirms the booking and a remaining balance settled in cash during the trip.

0. Offer a verified student status that grants an automatic discount on eligible trips.
0. Ensure security, scalability, and maintainability throughout the architecture.

1.2.6 Proposed Solution

ForsaDrive is designed as an integrated ecosystem composed of three coordinated components : a web application, a mobile application, and a shared backend. The platform supports the complete carpooling lifecycle, from registration to post-trip evaluation. A driver can publish a trip with its origin, destination, date, price, and number of available seats. A passenger can search for trips, filter results, consult the profile of the driver, book one or several seats, and confirm the booking through a fifty-percent prepayment. After the trip, both parties can rate each other, and any incident can be reported to the administration through a dedicated complaint channel.

The originality of the platform lies in the careful adaptation of every feature to the local context : the payment flow combines an online prepayment with a cash settlement, in line with the way Tunisian users currently handle informal carpooling ; the student status is verified through documents or a university email, which makes the discount both accessible and auditable ; and the interface is designed to support the languages effectively used by the target population. These choices, which are detailed in the following chapters, are what distinguish ForsaDrive from the international platforms that have so far failed to gain a strong foothold in the country.

Conclusion

This chapter introduced the host organization, presented the structural limits of the current transportation situation in Tunisia, and stated the motivations that led to the development of ForsaDrive. It also formulated the general and specific objectives of the project and outlined the proposed solution at a high level. The next chapter translates this general framework into a precise specification : it analyzes the existing competing solutions in more detail, formalizes the actors of the system, and lists the functional and non-functional requirements that the platform must fulfill.

Chapitre 2

Analysis and Specification of Requirements

Plan

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2 Development Methodology . . . 14

3 Identification of Actors . . . 16

4 Functional Requirements . . . 16

5 Non-Functional Requirements . . . 17

6 Global Use Case Diagram . . . 18

Introduction

The previous chapter established the context of the project and stated the high-level objectives of ForsaDrive. The present chapter takes the analysis one step further by clarifying, in a structured way, who the platform is for, what it must do, and what quality attributes it must respect. The reasoning starts with a comparative review of the mobility solutions on which Tunisians currently rely, ranging from informal social-media groups to international ride-hailing applications. It continues with the description of the development methodology adopted throughout the internship, the identification of the actors of the system, and the listing of the functional and non-functional requirements. The chapter closes with a global use case diagram that summarizes the scope of the platform and prepares the conception phase presented in the next chapter.

2.1 Study of Existing Solutions

Before designing ForsaDrive, it was essential to understand how Tunisians currently organize shared trips, both through formal applications and through informal channels. This step served two complementary purposes : it allowed an objective measure of the gaps left by existing tools and it informed several design decisions presented later, such as the combination of online prepayment and cash settlement.

2.1.1 Informal Carpooling through Social Media

A large share of carpooling activity in Tunisia takes place outside any structured platform. Facebook groups such as *Covoiturage Tunisie*, *Covoiturage Tunis–Sfax*, and regional groups dedicated to specific universities are among the most active channels. Trips are published as simple posts, and passengers contact drivers through private messages or phone calls.
While this informal system genuinely meets part of the demand, it suffers from several structural weaknesses :

- There is no verification of the identity of the driver or of the vehicle.
- Payments are handled entirely offline, with no trace of the transaction.
- Trip information is scattered across several groups, and no real filtering or sorting is possible.
- Cancellations and no-shows are frequent, since neither party has any concrete commitment.
- There is no rating system, which forces passengers to rely solely on the comments left under each post.

This mode of organization is tolerated rather than regulated, and its very popularity illustrates how strong the demand is for a structured digital alternative.

Figure 2.1 : Example of an informal carpooling post on a Facebook group

2.1.2 International Ride-Hailing Applications

Several international ride-hailing applications are currently active on the Tunisian market. Although they are not carpooling platforms in the strict sense, they are often mentioned by users as alternatives for point-to-point travel, which justifies their inclusion in this comparative study.

2.1.2.1 inDrive

inDrive is built on a fare-negotiation model : the passenger proposes a price and drivers can accept the offer or counter-offer. The application is mainly used inside cities such as Tunis, Sousse, and Sfax. However, inDrive is not designed for long-distance travel. Its pricing logic becomes impractical for inter-city routes, and drivers have very little incentive to accept negotiated long trips at a price they cannot adjust based on the number of passengers sharing the vehicle.

Figure 2.2 : inDrive logo

2.1.2.2 Bolt

Bolt operates as a classic ride-hailing service with fixed, algorithmic pricing. It is widely used for short urban trips, but it offers no mechanism for sharing a vehicle between passengers heading in the same direction, and its fares rise sharply on longer journeys, which excludes inter-city carpooling from its use cases.

Figure 2.3 : Bolt logo

2.1.2.3 Uber

Uber has only a limited presence in Tunisia and is essentially absent from the inter-city segment. For the use cases targeted by ForsaDrive, it is therefore not a realistic alternative.

2.1.3 International Carpooling Platforms

BlaBlaCar is the most well-known carpooling platform internationally and is also used, although on a modest scale, in Tunisia. It offers a structured experience built around verified profiles, integrated payments, and ratings. Its adoption in Tunisia, however, remains low for reasons that have little to do with the quality of the platform itself : payment methods are mostly limited to international cards, customer support operates in foreign languages, and the local user base is too small to ensure frequent matches on the most requested national routes. As a result, the value of the network — which is the main strength of carpooling platforms — is significantly reduced in the Tunisian context.

Figure 2.4 : BlaBlaCar logo

2.1.4 Comparative Table

Table 2.1 summarizes the main characteristics of the solutions reviewed above and contrasts them with the features targeted by ForsaDrive.

Tableau 2.1 : Comparison of existing solutions with ForsaDrive

Criterion	**Facebook**	**inDrive**	**Bolt**	**BlaBlaCar**	**ForsaDrive**
Identity verification	No	Partial	Yes	Yes	Yes
In-app payment	No	No	Yes	Yes	Yes
Long-distance rides	Yes	No	No	Yes	Yes
Rating system	No	Yes	Yes	Yes	Yes
Adapted to local context	Yes	Partial	Partial	No	Yes
Student discount	No	No	No	No	Yes
Wallet-based balance	No	No	No	No	Yes

The analysis shows that no existing option simultaneously offers the local fit of informal channels and the reliability of structured platforms. ForsaDrive is positioned precisely in this gap, combining the familiarity of practices that Tunisian users already adopt with the structure, security, and trust mechanisms that international platforms have demonstrated elsewhere.

2.2 Development Methodology

2.2.1 Choice of an Agile Approach

Given the scope of the project, which combines a web application, a mobile application, and a shared backend, a sequential approach such as the waterfall model was rapidly ruled out. Several reasons supported this decision. First, the requirements were expected to evolve as the design progressed, which is typical for a platform whose features have direct counterparts in the daily life of the target users. Second, the project required iterative feedback with the supervising team at ATOMIC IT in order to validate design choices step by step. Third, the team was small and the schedule short, two conditions under which the lightness of an agile process is a clear advantage over the heavier ceremonies of more formal methodologies. The **Scrum** framework was therefore selected as the development methodology of the project.

2.2.2 The Scrum Framework

Scrum organizes the work into short iterations called sprints, each of which produces a usable increment of the product. Every sprint is planned at the beginning, reviewed at the end, and followed by a short retrospective whose role is to identify what worked, what did not, and which adjustments to apply during the next iteration. Three roles structure the process : the *Product Owner*, who represents the client and maintains the prioritized backlog ; the *Scrum Master*, who facilitates the process and removes blocking issues ; and the *Development Team*, which builds the increment.

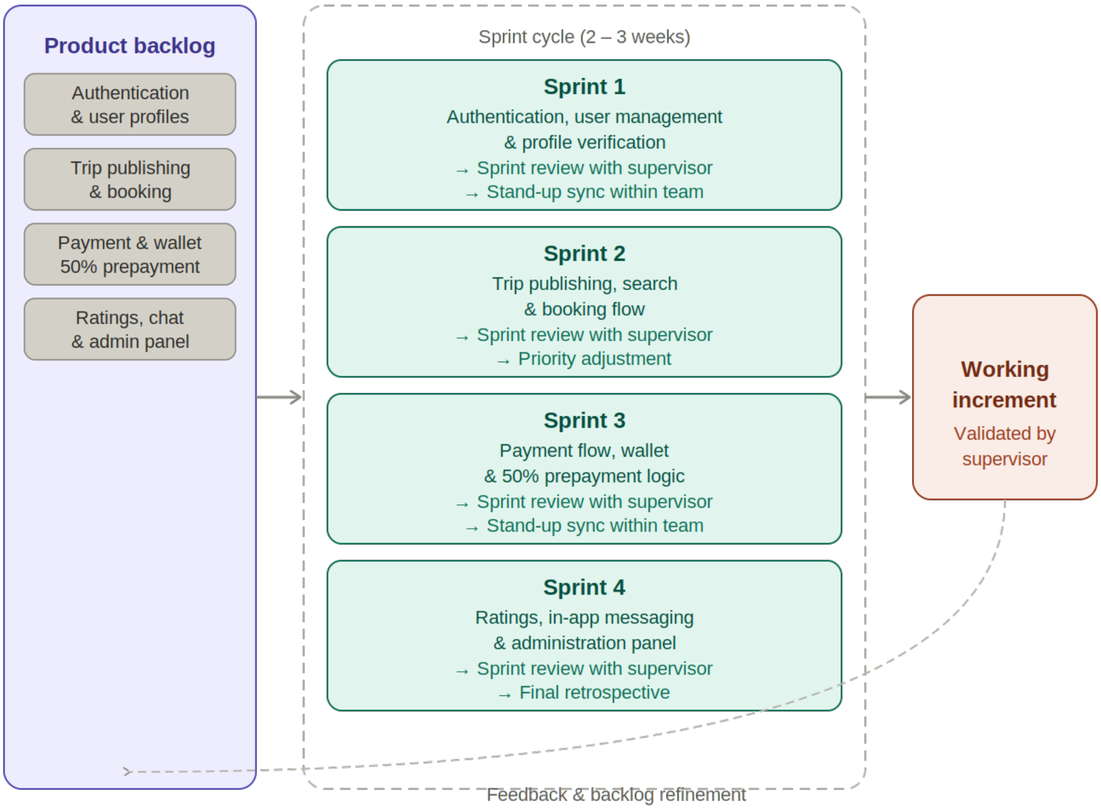

Figure 2.5 : Overview of the Scrum framework

2.2.3 Application to ForsaDrive

In the context of this internship, Scrum was applied in a simplified form adapted to a two-student team. The work was organized into four functional sprints, each of which produced a coherent increment of the platform.

0. **Sprint 1 — Foundations.** Project setup, authentication, profile management, role management, student verification, driver profile creation, and administrative validation.
0. **Sprint 2 — Rides and Bookings.** Ride publishing, search and filtering, single and group bookings, and management of booking requests.
0. **Sprint 3 — Payments and Intelligence.** Wallet, fifty-percent prepayment, student discount calculation, ride boosting, driver analytics, and recommendation logic.
0. **Sprint 4 — Community and Finalization.** In-app messaging, ratings, complaint handling, social feed, HelpDesk, multilingual support, testing, and deployment.

Short stand-up meetings were held inside the team to coordinate the day-to-day work, and weekly reviews were organized with the supervising engineer to validate progress and adjust priorities. This rhythm proved well suited to a project of this size, since it kept the team focused on a small set of features at a time while preserving the flexibility needed to absorb changes coming from the reviews.

2.3 Identification of Actors

In the UML sense, an actor is any entity, human or automated, that interacts with the system. Four human actors and one automated actor were identified for ForsaDrive.

Guest. An unauthenticated visitor who can browse the public landing page, consult general information about the platform, and start the registration process.

Passenger. A registered user looking for a trip. A passenger can search for available rides, book seats, pay the prepayment, communicate with the driver, and rate the trip after its completion.

Driver. A registered user who offers trips. A driver can create, modify, or cancel rides, accept or decline booking requests, and receive payments through the platform.

Administrator. The operator of the platform. The administrator is responsible for verifying user identities, validating student documents, moderating reported content, handling disputes, and supervising the overall activity of the platform.

System. An automated actor that triggers operations such as the calculation of the student discount, the dispatch of notifications, and the recomputation of the reliability score after each completed trip.

The role of *Student* is not modeled as a separate actor but as a status attached to a passenger account. This status is granted after a successful verification process and gives access to the fifty-percent discount on eligible trips.

2.4 Functional Requirements

Functional requirements describe what the system must do. They are grouped here by actor in order to reflect the structure of the use case diagram presented at the end of the chapter.

2.4.1 Requirements for the Guest

- Browse the public landing page and consult general information about the platform.
- Register a new account, choosing between a passenger and a driver role.
- Authenticate through email and password.
- Recover a forgotten password through a secure procedure.

2.4.2 Requirements for the Passenger

- Search for trips by departure city, arrival city, and date.

— Filter results by price, departure time, and driver rating.
— Consult the full details of a trip, including the profile of the driver.
— Reserve one or several seats on a trip.
— Pay fifty percent of the amount as a prepayment at the moment of booking.
— Benefit from the student discount once the account is verified as such.
— Cancel a booking under the cancellation rules defined by the platform.
— Rate and review a driver after a completed trip.
— Exchange messages with a driver through the in-app chat.

2.4.3 Requirements for the Driver

— Publish a new trip with its origin, destination, date, time, price, and number of available seats.
— Modify or cancel an existing trip while respecting the rules that protect already booked passengers.
— View and manage incoming booking requests.
— Access a wallet that summarizes earned, pending, and refunded amounts.
— Receive notifications about new bookings and incoming messages.
— Rate a passenger after a completed trip.

2.4.4 Requirements for the Administrator

— Authenticate through a dedicated administration panel.
— Manage user accounts (activation, deactivation, verification).
— Review and validate identity documents and student cards.
— Moderate reported trips, users, or messages.
— Consult global statistics on users, trips, and payments.

2.5 Non-Functional Requirements

Non-functional requirements describe the quality attributes that the system must respect. They constrain the design as much as the functional requirements do, since they directly affect the trust that users place in the platform.

Security. Authentication relies on hashed passwords. Sensitive operations such as payment and account modification require an active and validated session. All exchanges between client and server are protected by HTTPS, and identity documents are stored in a dedicated, access-controlled location.

Performance. The system must remain responsive under normal load. Search results, in particular, must be returned within a delay short enough to preserve a fluid user experience.

Availability. The platform must remain reachable at all times, except during scheduled maintenance windows announced in advance to the users.

Usability. The interfaces are clear and consistent across the web and mobile platforms. A new user must be able to book a trip without external assistance, which requires careful attention to information density, form validation, and feedback messages.

Portability. The mobile application runs on recent versions of Android and iOS. The web application is compatible with the major modern browsers.

Scalability. The architecture allows new features to be added without rewriting existing modules. This is particularly important for ForsaDrive, since the roadmap includes intelligent services that will be plugged into the existing booking workflow.

Maintainability. The source code follows clear conventions and a layered organization, so that a new contributor can become productive within a reasonable amount of time.

2.6 Global Use Case Diagram

The global use case diagram synthesizes the main interactions between the actors and the system. It provides the first complete view of the functional coverage of ForsaDrive and serves as the entry point for the conception phase of the next chapter.

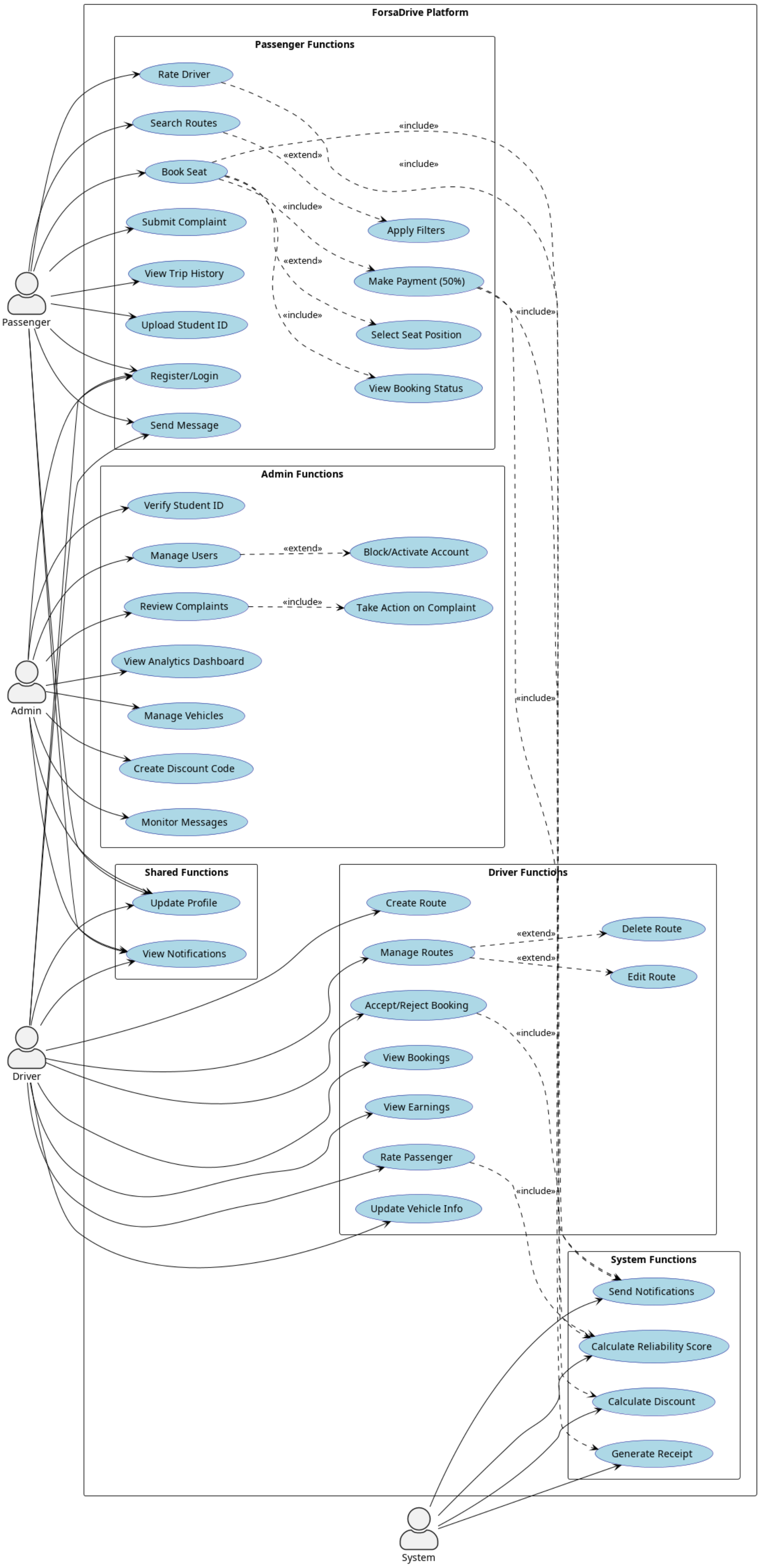

FIGURE 2.6 : Global use case diagram of ForsaDrive

Conclusion

This chapter defined the functional scope of ForsaDrive. The comparative study of existing solutions confirmed the absence of a carpooling platform genuinely adapted to the Tunisian context and clarified the gap that the project aims to fill. The presentation of the Scrum methodology and its concrete application to a four-sprint plan introduced the rhythm under which the work was carried out. The identification of the actors and the listing of the functional and non-functional requirements then translated the high-level objectives of the project into a precise specification, which is summarized in the global use case diagram. The next chapter takes this specification as input and produces a detailed conception of the platform, covering its architecture, the dynamic behavior of its most representative scenarios, and the structure of its data.

Chapitre 3

Conception

Plan

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Introduction

Once the requirements of a platform have been clarified, the conception phase becomes the bridge between what the system should do and how it will actually be built. This chapter presents the conception of ForsaDrive at progressively finer levels of detail. It opens with the general software architecture, then refines the most representative use cases identified in the previous chapter, illustrates the dynamic behavior of the system through sequence and activity diagrams, and finally describes the structural backbone of the application by means of a class diagram and a relational schema. Throughout the chapter, the modeling choices are justified in light of the constraints of the Tunisian context : a user base that is more comfortable with mobile devices than with web browsers, payments that are still partly handled in cash, and a market in which trust between drivers and passengers must be built feature by feature.

3.1 General Architecture

3.1.1 Client–Server Architecture

ForsaDrive is organized around a classic client–server architecture. A unified REST API is exposed by a backend server and is consumed by two distinct clients : a web application targeted at administrators and at desktop users, and a mobile application that covers the daily usage of drivers and passengers. This separation has two main benefits. It allows the two interfaces to evolve at their own pace, as long as the API contract remains stable, and it avoids duplicating the business logic on the client side, where it would be both harder to maintain and easier to bypass. External services such as the payment gateway, the notification service, and the file storage that hosts student identity documents are reached exclusively through the backend, so that no sensitive credential is ever exposed on a client device.

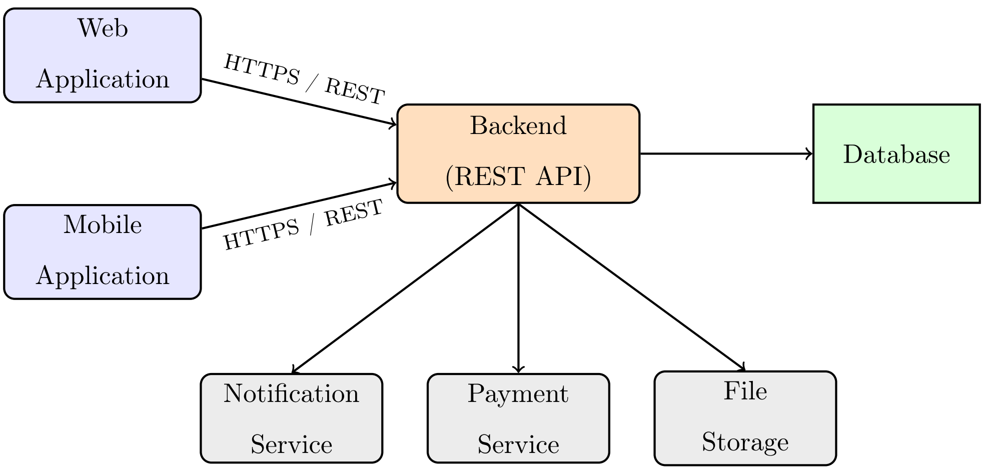

Figure 3.1 : Client–server architecture of ForsaDrive

3.1.2 Three-Tier Organization

Internally, the backend follows a three-tier organization that separates presentation, business logic, and data persistence. This decomposition reflects a deliberate effort to isolate the parts of the system that change at different rhythms : the presentation tier evolves with the user interface, the application tier with the business rules, and the data tier with the persistence model.

Presentation tier. Composed of the web and mobile applications, this layer is responsible for user interaction, form validation on the client side, and rendering. It never accesses the database directly and only communicates with the application tier through the REST API.

Application tier. Contains the business rules of ForsaDrive : route publication, booking logic, prepayment computation, application of the student discount, ride lifecycle, ratings and reliability scoring, and complaint handling. It is exposed through a REST API secured by JSON Web Tokens.

Data tier. Stores persistent information such as users, routes, bookings, payments, ratings, and messages. It is accessed exclusively through the application tier, which guarantees the integrity of the business rules at all times.

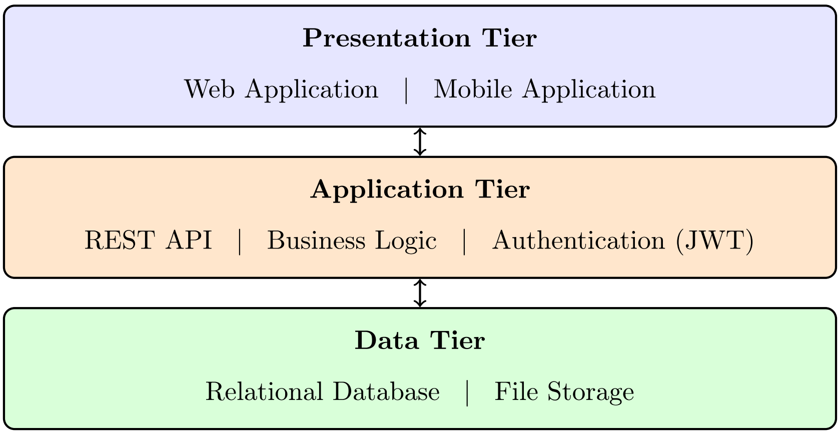

FIGURE 3.2 : Three-tier organization of the platform

3.1.3 External Services

In addition to its own three tiers, the backend interacts with three external building blocks. The **payment service** processes the fifty-percent prepayment that confirms a booking ; in the first version of ForsaDrive, this service is simulated, but the architecture is designed so that it can be replaced by a real provider without any impact on the rest of the code. The **notification service** delivers booking, payment, and message alerts to drivers and passengers, both through push notifications on the mobile application and through email when relevant. Finally, a dedicated **file storage** keeps student identity documents and profile pictures, so that the relational database is not polluted with binary content and so that documents can be served through short-lived signed URLs that protect them from unauthorized access.

3.1.4 Security Mechanisms

Authentication relies on signed tokens delivered after a successful login. Every protected endpoint validates the token, extracts the user identity, and checks the associated role before executing the requested operation. Passwords are hashed with a modern algorithm and are never stored or transmitted in clear. All communications go through HTTPS, and sensitive operations such as payment, account suspension, or student verification are logged on the server side for audit purposes. Identity documents stored in the file service are accessible only to authenticated administrators, through signed URLs that expire after a short delay. Together, these mechanisms reduce the attack surface of the platform and align its handling of personal data with the expectations of users entrusting their identity documents to the system.

3.2 Refinement of Use Cases

The global use case diagram of ForsaDrive, presented in Figure 3.3, regroups the functionalities of the platform around four actors – the *Passenger*, the *Driver*, the *Admin*, and the *System* itself, the latter being responsible for automated operations such as discount calculation, reliability scoring, and notification dispatching. Each actor is restricted to a coherent set of use cases, and a few of them are connected through `«include»` or `«extend»` relationships when an operation must trigger an automated step or when a behavior is optional.

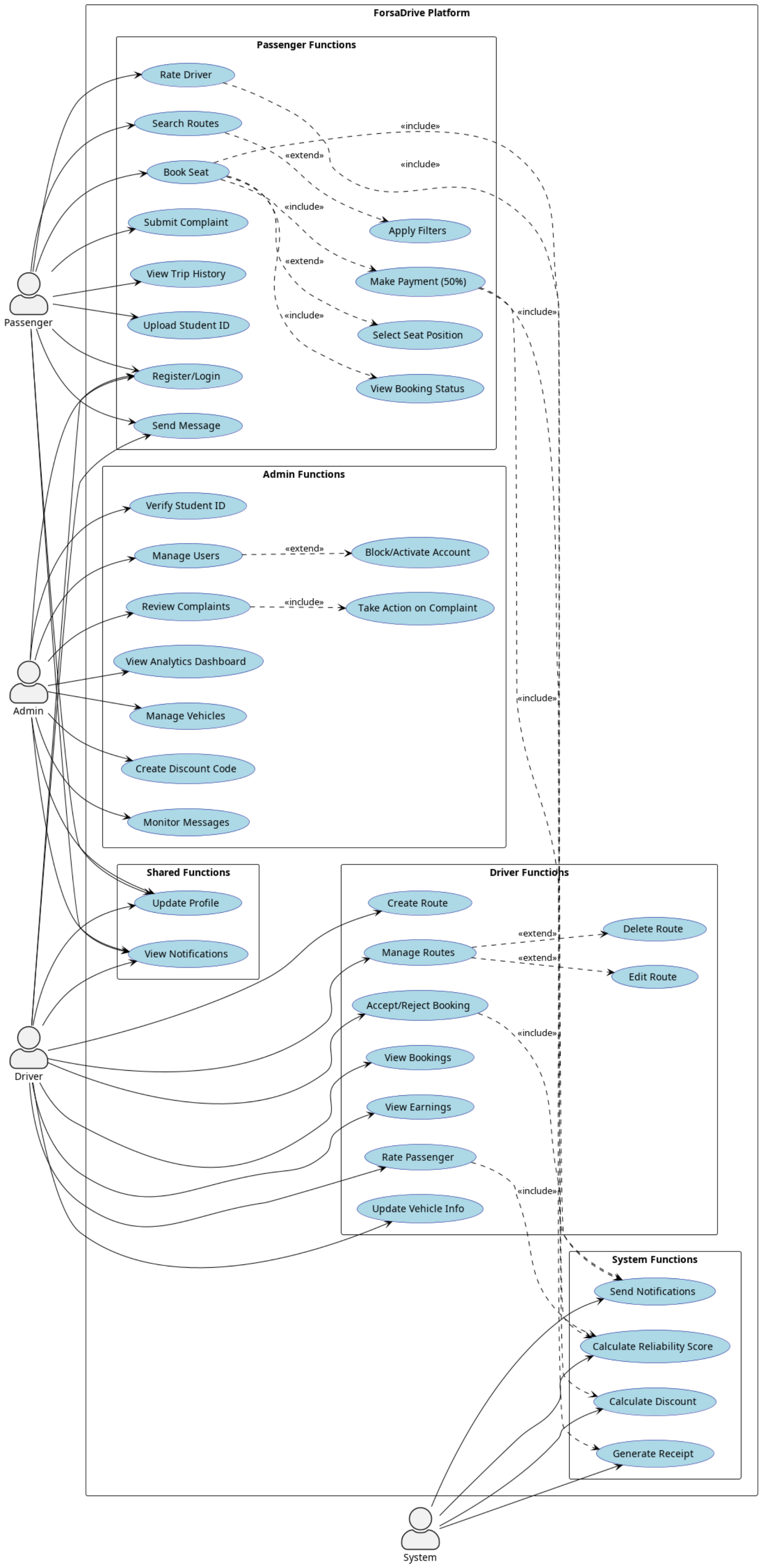

FIGURE 3.3 : Global use case diagram of ForsaDrive

The most representative use cases are refined below in textual form, in order to expose their internal logic and the alternative scenarios that the system must handle.

3.2.1 Use Case : Publish a Route

Actor : Driver.

Precondition : The driver is authenticated, the account is active, and a vehicle has been registered on the profile.

Main scenario :

0. The driver selects the action of creating a new route.
0. The system displays the route creation form.
0. The driver fills in the origin, the destination, the departure date and time, the price per seat, and the number of available seats.
0. The driver confirms the creation.
0. The system validates the data, persists the route with the status *ACTIVE*, and notifies the passengers whose saved searches match the new offer.

Alternative scenarios :

— If a mandatory field is missing or invalid, the system displays an error message and the route is not created.

— If the driver account is suspended, the system rejects the request with an explanatory message.

— If no vehicle is registered yet, the driver is redirected to the vehicle registration page.

3.2.2 Use Case : Book a Ride

Actor : Passenger.

Precondition : The passenger is authenticated and at least one seat is available on the selected route.

Main scenario :

0. The passenger searches for routes by entering origin, destination, and date.
0. The system returns the matching routes and applies the student discount on the displayed price if the passenger is a verified student.
0. The passenger opens the details of a route and selects the number of seats and, when available, a seat position.
0. The system computes the total amount and the fifty-percent prepayment.
0. The passenger confirms the booking and the prepayment.
0. The system processes the prepayment, registers the booking with the status *PENDING*, and notifies the driver.

Alternative scenarios :

— If the prepayment fails, the booking is not created and the passenger is invited to retry or to cancel.
— If the seat becomes unavailable between the selection and the confirmation, the booking is rejected and the passenger is informed.
— If the driver later refuses the booking, the prepayment is automatically refunded.

3.2.3 Use Case : Verify Student Status

Actor : Admin.

Precondition : The passenger has uploaded a student identity document and a verification request is pending.

Main scenario :

0. The admin opens the panel of pending student verifications.
0. The admin selects a request and reviews the uploaded document, the declared university, and the email domain associated with the account.
0. The admin approves the request.
0. The system updates the verification status to *APPROVED*, marks the user as a verified student, and sends a notification to the passenger.

Alternative scenarios :

— If the document is unreadable, expired, or does not match the declared university, the admin rejects the request and provides an explicit reason. The student receives a notification and is allowed to resubmit a corrected document without restarting the entire process.

3.2.4 Use Case : Rate a Trip

Actor : Passenger or Driver.

Precondition : The trip is marked as completed.

Main scenario :

0. Once the trip is completed, the system invites both parties to rate each other.
0. The user selects a score between one and five and optionally writes a short comment.
0. The system stores the rating, updates the average score of the rated party, and recomputes the reliability score of the driver when applicable.

3.3 Sequence Diagrams

Sequence diagrams describe, over time, the messages exchanged between the actors and the components of the system for a specific scenario. Three critical scenarios of ForsaDrive are detailed below : the creation of a route by a driver, the booking of a ride by a passenger with prepayment, and the verification of a student account by an administrator.

3.3.1 Sequence Diagram : Driver Creates a Route

The diagram in Figure 3.4 shows how a driver publishes a new route. The frontend first performs a client-side validation in order to catch the most common mistakes early ; the request is then sent to the backend, which checks the eligibility of the driver and the presence of a registered vehicle before persisting the route. Once the route is saved, the notification service informs the passengers whose saved searches match the new offer, which closes the loop between supply and demand without requiring any extra action from the driver.

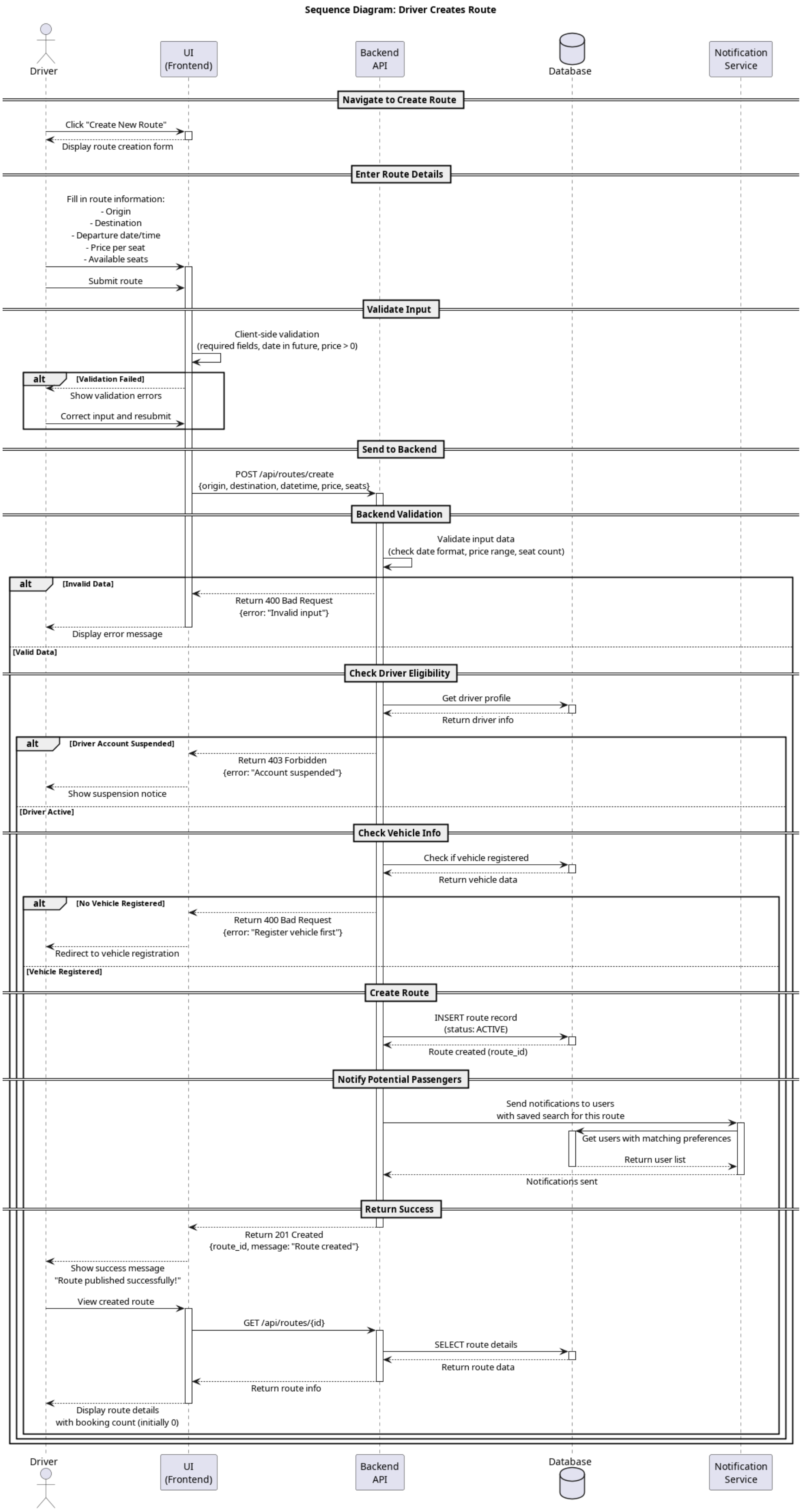

FIGURE 3.4 : Sequence diagram of the publication of a route by a driver

3.3.2 Sequence Diagram : Passenger Books a Ride

Figure 3.5 highlights the interaction between the passenger, the application, the payment service, the notification service, and the database during a booking. The student discount is applied at the price computation step, before the prepayment of fifty percent is processed. The booking is only persisted if the prepayment succeeds, which guarantees the commitment of the passenger and protects the driver from no-shows – a problem that, as discussed in the previous chapter, undermines informal carpooling in Tunisia.

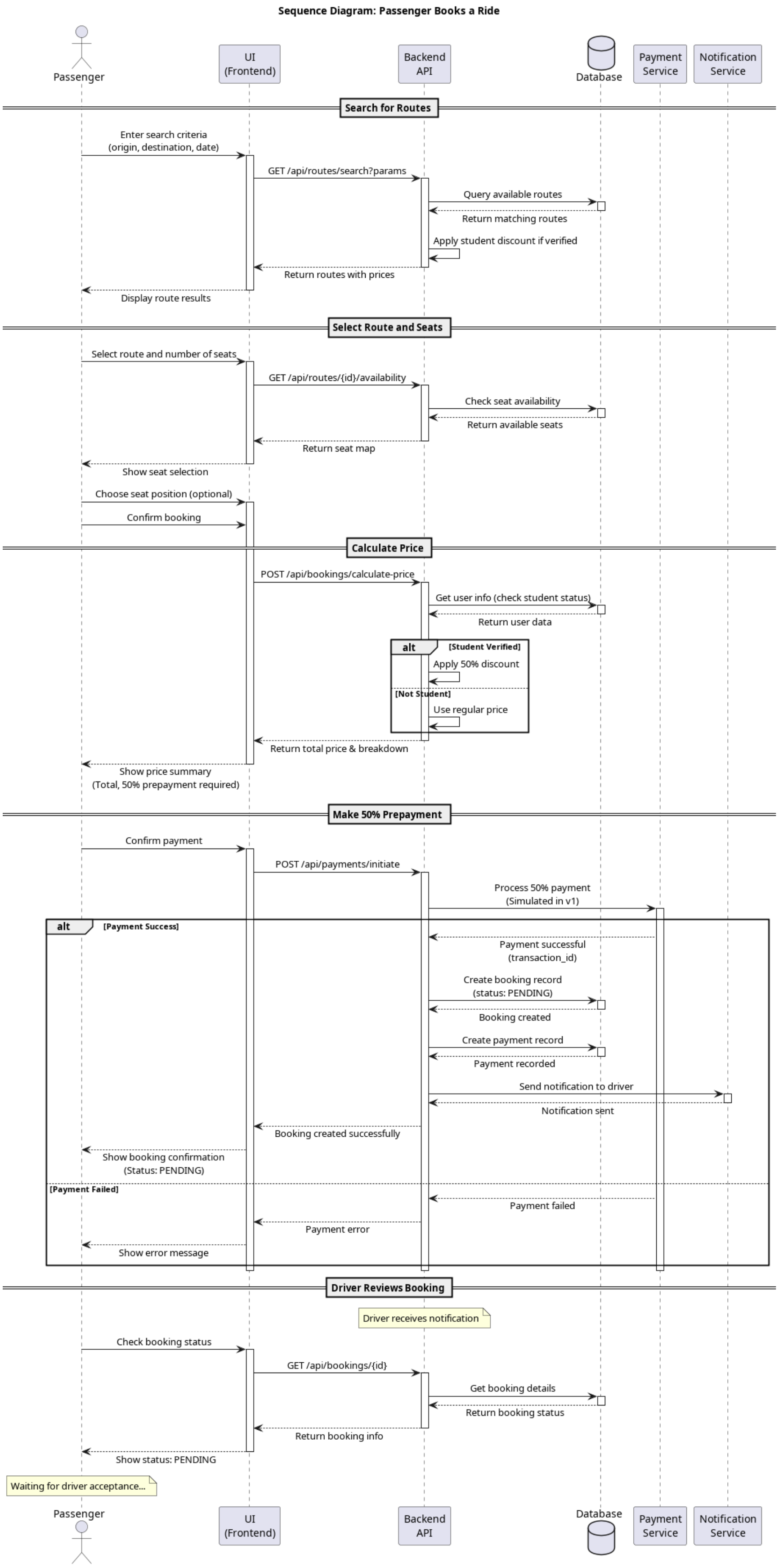

Figure 3.5 : Sequence diagram of a booking with prepayment

3.3.3 Sequence Diagram : Admin Verifies a Student

The verification of a student account is a sensitive operation, since it conditions the application of the discount and therefore the financial integrity of the platform. The diagram in Figure 3.6 describes the full flow, from the consultation of the pending requests to the dispatch of the approval or rejection notification. In case of rejection, an explicit reason is stored, so that the student can resubmit a corrected document without having to start the process from scratch.

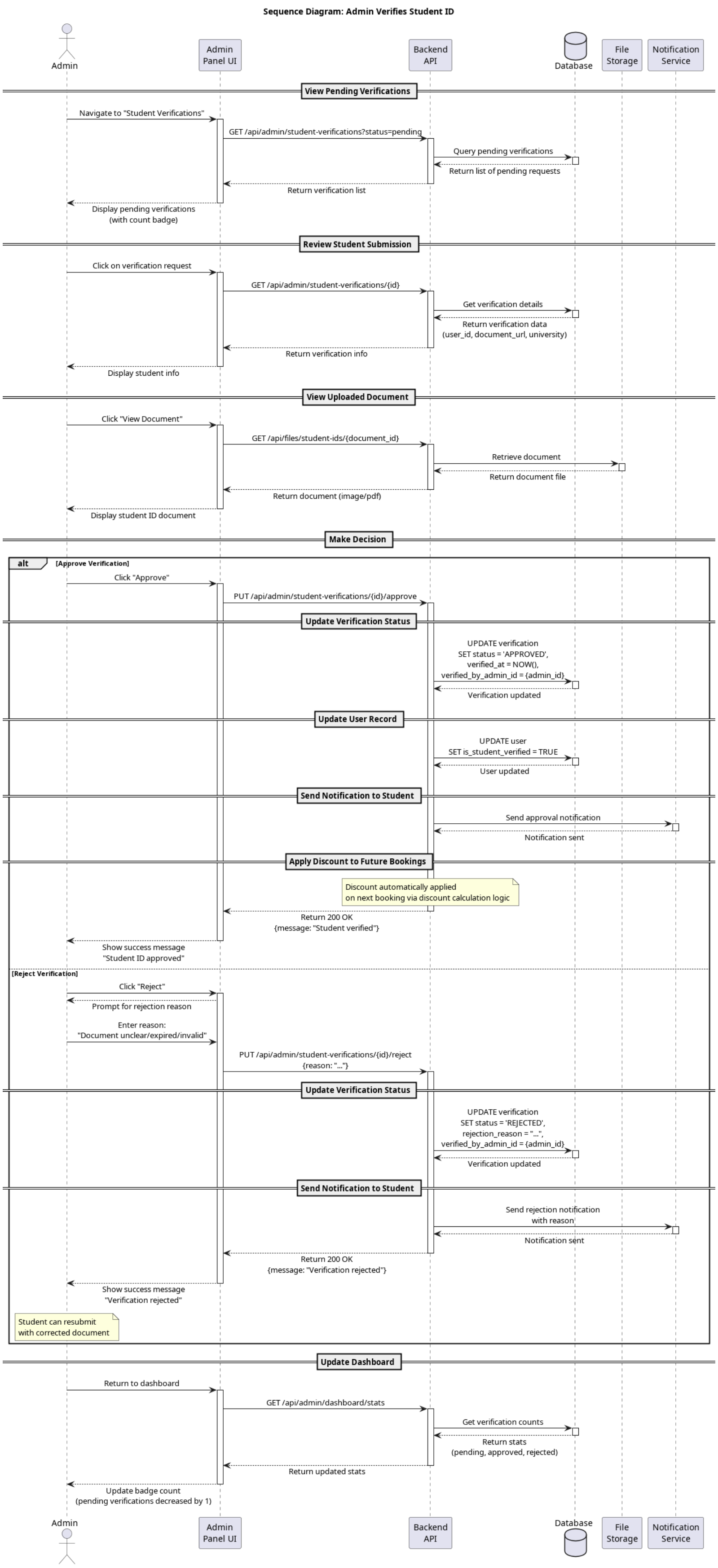

FIGURE 3.6 : Sequence diagram of the student verification by an administrator

3.4 Activity Diagram of the Booking Flow

In order to give a global view of the booking process, an activity diagram is used to follow the journey of a passenger from the moment the application is opened until the trip is rated. The diagram, shown in Figure 3.7, is split into three swimlanes that correspond to the passenger, the system, and the driver. It complements the sequence diagrams by emphasizing the decision points – the retry of a failed payment, the acceptance or refusal of the booking by the driver, the optional seat selection – rather than the technical messages exchanged between the components.

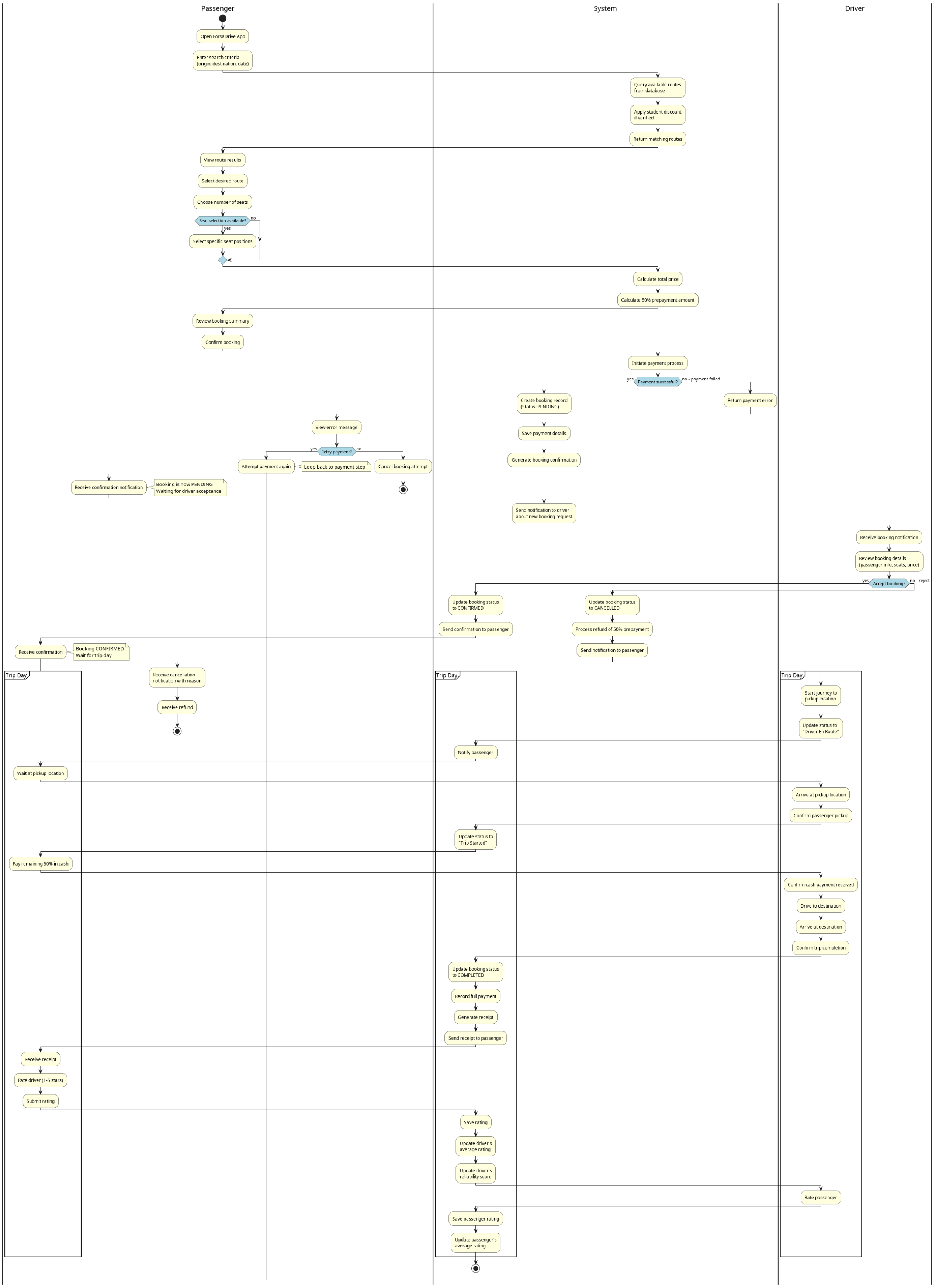

FIGURE 3.7 : Activity diagram of the end-to-end booking flow

3.5 Class Diagram

The class diagram describes the structural backbone of the application. Given the breadth of the platform, the diagram regroups around twenty classes that cover not only the core booking entities but also the secondary features that have been identified as relevant for a Tunisian carpooling platform : student verification, complaints, ratings, route boosting, social feed, and analytics. The full diagram is shown in Figure 3.8.

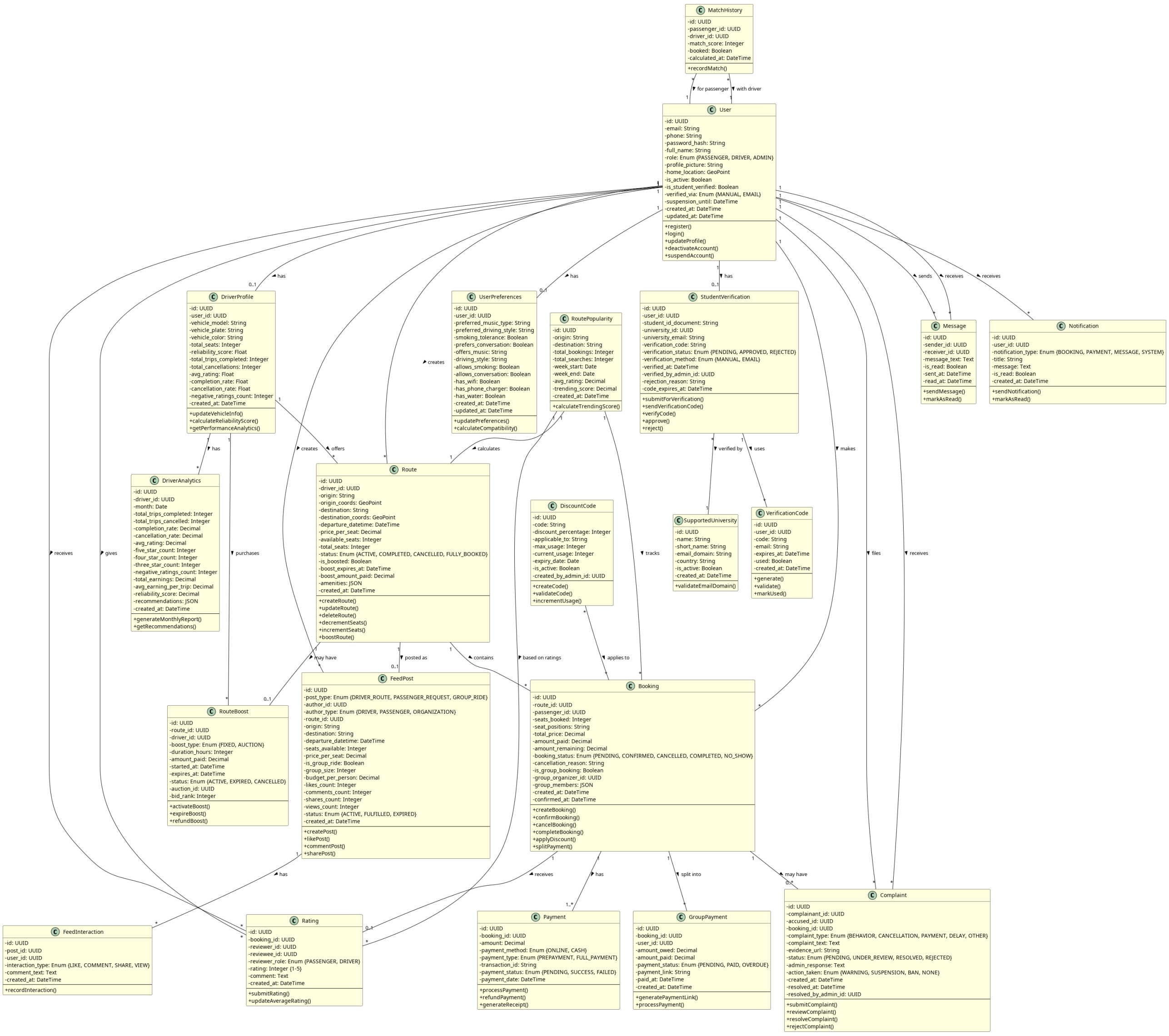

FIGURE 3.8 : Class diagram of ForsaDrive

For readability, the main classes are organized into five logical groups.

Users and profiles. The class *User* represents any registered account and centralizes the credentials, the personal information, the verification status, and the role (*PASSENGER*, *DRIVER*, or *ADMIN*). It is enriched by *DriverProfile*, which holds the vehicle information and the reliability indicators of a driver, and by *UserPreferences*, which describes the preferred travel conditions of a user (music, conversation, smoking tolerance, on-board amenities).

Trips and bookings. The class *Route* represents a ride offered by a driver, with its origin, its destination, its date, its price, and its available seats. *Booking* represents a reservation made by a passenger on a route and stores the booked seats, the total amount, the prepayment, and the booking status. The optional class *GroupPayment* models the splitting of a booking among several passengers traveling together.

Payments and ratings. The class *Payment* represents the financial operations linked to a booking, whether it is the prepayment or the final settlement. *Rating* represents the feedback left by a user after a completed trip and feeds back into the average score and the reliability indicators stored on *DriverProfile*.

Verification and discounts. The class *StudentVerification* stores a verification request, with the uploaded document, the declared university, and the decision of the administrator. It is supported by *SupportedUniversity*, which lists the institutions whose email domains are recognized by the platform, and by *VerificationCode*, which manages email verification codes. The class *DiscountCode* models promotional codes that can be applied at booking time.

Communication and moderation. The classes *Message* and *Notification* carry the conversations between users and the alerts emitted by the system. *Complaint* allows a user to report a problem after a trip and gives the administrator the means to take action, ranging from a warning to a permanent ban.
Around these core classes, complementary entities such as *DriverAnalytics*, *RoutePopularity*, *MatchHistory*, *RouteBoost*, *FeedPost*, and *FeedInteraction* support the analytical and social dimension of the platform without polluting the central booking workflow.

3.6 Database Design

The class diagram has been translated into a relational model that serves as the schema of the backend database. The translation follows the standard rules : each persistent class becomes a table, each one-to-many association becomes a foreign key, and many-to-many associations are realized through dedicated join tables.

3.6.1 Logical Schema

The logical schema below uses the conventional notation, where primary keys are underlined and foreign keys are marked with the # symbol. For the sake of readability, only the most representative tables are listed.

- **User** (id_user, full_name, email, phone, password_hash, role, profile_picture, is_active, is_student_verified created_at, updated_at)
- **DriverProfile** (id_driver_profile, #id_user, vehicle_model, vehicle_plate, vehicle_color, total_seats, avg_rating, reliability_score, completion_rate, cancellation_rate, created_at)
- **UserPreferences** (id_preferences, #id_user, preferred_music_type, prefers_conversation, smoking_tolera has_wifi, has_phone_charger)
- **Route** (id_route, #id_driver, origin, destination, departure_datetime, price_per_seat, total_seats, available_seats, status, is_boosted, created_at)
- **Booking** (id_booking, #id_route, #id_passenger, seats_booked, seat_positions, total_price, amount_paid, amount_remaining, booking_status, created_at, confirmed_at)
- **Payment** (id_payment, #id_booking, amount, payment_method, payment_type, transaction_id, payment_status, payment_date)
- **Rating** (id_rating, #id_booking, #id_reviewer, reviewer_role, score, comment, created_at)
- **StudentVerification** (id_verification, #id_user, #id_university, student_id_document, university_email, verification_status, verified_at, #id_admin, rejection_reason)
- **SupportedUniversity** (id_university, name, short_name, email_domain, country, is_active)
- **DiscountCode** (id_discount, code, discount_percentage, applicable_to, max_usage, current_usage, expiry_date, is_active, #id_admin)
- **Message** (id_message, #id_sender, #id_receiver, message_text, is_read, sent_at, read_at)
- **Notification** (id_notification, #id_user, notification_type, title, message, is_read, created_at)
- **Complaint** (id_complaint, #id_complainant, #id_accused, #id_booking, complaint_type, complaint_text, evidence_url, status, action_taken, created_at, resolved_at, #id_admin)

3.6.2 Constraints and Indexing

A few business rules deserve to be enforced at the database level rather than only in the application code, since this guarantees their consistency even when the data is touched outside the normal application paths. The number of available seats on a route is constrained between zero and the total number of seats. The booking status follows a controlled state machine (*PENDING*, *CONFIRMED*, *CANCELLED*, *COMPLETED*, *NO_SHOW*), and the corresponding transitions are checked before any update. Indexes are added on the columns that are frequently used for filtering, in particular the origin and the destination of a route, the departure date, and the foreign keys that link bookings to their route and to their passenger. These indexes ensure that the search operation, which is the most frequent action on the platform, remains responsive even as the volume of data grows.

Conclusion

This chapter presented the design of ForsaDrive at several levels of abstraction. The general architecture defined the boundary between the clients, the backend, and the external services, and clarified the security mechanisms that protect the platform. The refined use cases, together with the sequence and activity diagrams, described the most sensitive scenarios of the application – namely the publication of a route, the booking of a ride with prepayment, and the verification of a student. Finally, the class diagram and the relational schema laid down the data foundations of the system. With this design in place, the next chapter turns to the implementation : the technologies that have been chosen, the structure of the source code, and the most representative screens of the resulting application.

Conclusion

To conclude, this Final Year Project was carried out within **ATOMIC IT** as part of the development of **ForsaDrive**, a Tunisian carpooling platform available as both a web and mobile application. This project allowed us to apply the theoretical knowledge acquired during our studies at the Higher Institute of Technological Studies of Kelibia to a real-world software development context.

Throughout this experience, we were able to work on designing and implementing a complete system that connects drivers and passengers in an efficient, secure, and user-friendly way. This included managing key functionalities such as trip creation, seat reservation, pricing flexibility, and a prepayment system, which helped us better understand real-world constraints in software engineering projects.

Working on ForsaDrive significantly improved our technical and analytical skills, particularly in web and mobile development, system design, and problem-solving. It also strengthened our understanding of user-centered design and the importance of building practical solutions adapted to the local Tunisian context.

This project was not only a technical achievement but also a valuable professional experience. Working within **ATOMIC IT** and under the supervision of **Mr. Khalil Selmi** (professional supervisor) and **Ms. Ines Ben Nasr** (academic supervisor) allowed us to benefit from continuous guidance, feedback, and mentorship throughout the development process.

Finally, this work marks an important step in our academic journey. It provided us with practical exposure to software engineering practices and prepared us for future professional opportunities in the field of information technology.

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Annexes

Annex A : Product Backlog Extract

Figure 4.1 : Extract of the product backlog used during the project

Annex B : REST API Endpoints Overview

Figure 4.2 : Overview of the main REST API endpoints of ForsaDrive

Annex C : Postman Test Collection

Figure 4.3 : Postman test collection used for integration testing

Annex D : Database Structure in MongoDB Compass

Figure 4.4 : Collections of the ForsaDrive database

ملخّص

هذا العمل يندرج في إطار مشروع ختم الدراسات لنيل الإجازة في تكنولوجيا المعلومات للسنة الجامعية ٥٢٠٢-٦٢٠٢. يتمثل المشروع في تطوير منصة نقل تشاركي تونسية تُدعى ْسرسَرِ ْ، وهي عبارة عن تطبيق ويب وتطبيق جوال يهدفان إلى تسهيل التنقل وتقليل التكاليف وتعزيز المشاركة بين المستخدمين.

Résumé

Ce travail s'inscrit dans le cadre du projet de fin d'études pour l'obtention d'une licence en technologie de l'information pour l'année universitaire 2025-2026. Le projet consiste à développer une plateforme tunisienne de covoiturage appelée "ForsaDrive", comprenant une application web et une application mobile visant à faciliter les déplacements, réduire les coûts et encourager le partage entre utilisateurs.

Abstract

This work is part of the Final Year Project for obtaining a bachelor's degree in Information Technology for the academic year 2025-2026. The project consists of developing a Tunisian carpooling platform called "ForsaDrive", including both a web application and a mobile application, aiming to facilitate transportation, reduce costs, and promote shared mobility among users.

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